CoringTreespotters

This is the WDC for the project involving the phenology data collected by the treespotters along with the cores that will be collected in April 2025. It will be included in Christophe's Chapter 2

Important: main and original repo is here

Last Modified:

29 October 2026

Date of Study

2015 to December 2025.

Date of Public Archiving:

Spring 2027 (estimated)

Goal

Understand how growth and growing season length relate for mature trees.

Contributors

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- Elizabeth Wolkovich
- Catherine Chamberlain
- Neil Pederson

General Files

File	Where	What
		Packing list of stuff we need to bring when

packingList.md	Github	we'll core trees in April 2025
individualphenometricsdatafield_descriptions.xlsx	Github	Metadata downladed from NPN
45-2024 Accepted Proposal to access the living collections for research purposes.pdf	Github	Signed proposal to core the trees at the arboretum
Wolkovich collections access letter 27 January 2025.pdf	Github	Acceptance letter to core the trees at the arboretum
airfareSpring2025.md	Github	Flight comparison for trip to Boston
fieldNotesP1.pdf	Github	Field note sheet for all trees to print out (page 1)
fieldNotesP2.pdf	Github	Field note sheet for all trees to print out (page 2)
howToGPS.md	Github	How to use the GPS and import file in it
treesToCoreHarvard.gpx	Github	GPX to input into the GPS
housingNotes.md	Github	Dana greenhouse instructions
SectionCLivingCollectionAccessed.xlsx	Github	Spreadsheet requested to be filled by the people at the Arboretum with notes on the tree condition and the number of cores collected.

dailylog.tex	Github	Quick daily log when Mao and I went to core trees at the arboretum
GeneralCoringImpactUpdate2021finalFromNeilP.docx	Github	impacts of coring. Shared by Mao, from Neil
lifeHistorySpecies.xlsx	Github	Notes on life history strategies of all our species. Scraped from Sylvics
notesCoreMeasuring.md	Github	Notes on core measuring cores with imageJ
2023v2.1USA-NPNPlantandAnimalPhenophase_Definitions.pdf	Github	NPN phenophase definitions. Most recent version as of the date of study
partialPoolingDecision.md	Github	Quick notes on why we decided to partial pool species or not
visualCrossDating.md	Github	Visual cross dating notes

Data

File	Where	What
observerations updated table.xlsx	Github	Data transfered from main repo on the number of observations we have for each individual tree.
		List that Micheal Dosmann (Curator of Arnold

2025ApprovedPlantListforcoring.xlsx	Github	Arboretum) provided us. These are the trees they accepted that we core.
listTreesfromInteractiveMap.csv	Github	Data download from the arboretum website
individualphenometricsdata2025.csv	Github	Recent data downloaded from NPN
weldclimate.csv	Github	Climate data from weld climate station
treeswithPhenodata.csv	Github	Getting the trees for which we have phenological observations. This file was sent to Michael Dosmann
2025ApprovedPlantListforcoring.csv	Github	Condition of the trees we cored. Note that Q. coccinea was a mistake and shouldn't be in there
cleanedtreeallcues.csv	Github	Transfered from main repo
Folder: cores	Github	All the csvs for the cores in inches. Need to be cleaned in the cleaning folder using the different codes in there
individualphenometricsdatafield_descriptions.xlsx	Github	metadata from individual phenometric data exported from NPN
individualphenometricsdata2025.csv	Github	Individual phenometric data from NPN
listTreesfromInteractiveMap.csv	Github	List of our trees from the interactive map from the Aboretum so we could have more information on our studied trees.

loganAirportClimate.csv	Github	Logan airport climate data
SlabCity7_BEPArw.txt	Github	Paper birch tree ring data from Harvard forest given from neil
SlabCity8_BEPArw.txt	Github	Paper birch tree ring data from Harvard forest given from neil
weldhillClimateCleaned.csv	Github	Weld hill climate cleaned in Wildchrokie cleaning folder

Code

Code	Where	What
2018Data.R	Github	Transfered from main repo. Code visualize the data.
Cleaning_data.R	Github	Transfered from main repo. Cleaning the data
cleanStatusIntesity.R	Github	Cleaning an old version of the data. Will probably be deleted.
cleanTS_2025.R	Github	**MAIN** code to clean all the treespotters data acquired from NPN
CuesandPheno.R	Github	Transfered from main repo.
ListTreesWithPhenodata.R	Github	Quick code to look at which tree individuals we have phenodata for
mapArboretum.R	Github	Map the treespotters trees
numberandfigures2019.R	Github	Transfered from main repo.
TS_stan.R	Github	Transfered from main repo.
		The file that sources all the codes

getAll.R	Github	in the source folder
calculateGrowingSeasonGDD.R	Github	Calculates GDD for the growing season
cleaningCores.R	Github	Takes csv cores in the input folder, converts the inches in mm and clean the files in general
cleanObsData.R	Github	From main repo. Cleans phenology observational data
Folder: cofecha	Github	Tried to make Cofecha working. Never worked out.
mcmcanalysistools_rstan.R	Github	Mike Betancourt's analyses tools
mcmcvisualizationtools.R	Github	Mike Betancourt's visualization tools
simOneSpp.stan	Github	Stan program to simulate data so we can determine what sample size is required to observe an effect.
TSmodelGrowthGDD_noPP.stan	Github	Stan program to test if PP pooling species is better
TSmodelGrowthGDD.stan	Github	Stan program of main model with GDD as a predictors
TSmodelGrowthGSL.stan	Github	Stan program of main model with GSL as a predictors
TSmodelGrowthSOS.stan	Github	Stan program of main model with SOS as a predictors
TSmodelGrowthEOS.stan	Github	Stan program of main model with EOS as a predictors
TSmodelGrowth_zEOSonSOS.stan	Github	Stan program to understand the effect of SOS on EOS
		Stan program to check the model

TSmodelGrowthGDD_largerPriors.stan	Github	with GDD as a predictor with larger priors.
TSmodelGrowthPreviousYear.stan	Github	Stan program for previous year model.
TSmodelGrowth_z.stan	Github	Stan program, generic z scored model
TSmodelGrowthGDD_noayear.stan	Github	Stan program, no intercept on year
TSmodelGrowthGDD10.stan	Github	Stan program, old version of gdd model, using 10 C as a base temperature for GDD calculations
TSmodelGrowthGDD5.stan	Github	Stan program, old version of gdd model, using 5 C as a base temperature for GDD calculations
TSgrowthModelsMain.R	Github	Main models code including all posterior calculations
TSeffectSizeSim.R	Github	Simulating sample size required to observe an effect given our very low sample size.
mapArboretum.R	Github	Code for plotting the trees on a map of the Arboretum
TSgrowthModelsMain_plots.R	Github	All plots from posterior calculations made in TSgrowthModelsMain.R and some empirical data plots (e.g. boxplots)
plottingEmpiricalData.R	Github	Empirical data visualization
TSmodelGrowthPreviousYear.R	Github	Previous year model posterior calculations
longTermGDD.R	Github	Calculating long term GDD for some allometry plots made in plottingEmpirical data.

TSclimate_plots.R	Github	Some climate plots that we never really used
bepaHarvardForest.R	Github	Bepa dendrochronology plots from Harvard forest
TSmodelGrowthGDD_GDD5VSGDD10.R	Github	Comparison of GDD calculations using 10 vs 5 C as a baseline
TSmodelGrowthGDD_AVGvsAbsolute.R	Github	Average across species GDD calculations vs absolute
simTS.R	Github	Old code to simulate data
coordforGPS.R	Github	Code to get the coordinates inside the GPS
ListTreesWithPhenodata.R	Github	Some code to select the trees with phenodata